DULUTH
3 NOV

Side Meeting: AI Agent Discovery

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For advice, please talk to Working Group chairs or Area Directors.

Agenda

Presentation
- Problem Space, Core Needs
- Solution Space, Core Features
- Example Solution: Diagram, Zonefile + RRs

Discussion
- IANA Considerations
- Registry vs &| Index
- Collaboration Opportunities

Q&A

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Problem Space

AI agents need scalable, decentralized, and distributed mechanisms to discover one another prior to agent-to-agent data and application exchanges.

Drafts:
- draft-mozley-aidiscovery-00
- draft-ye-problems-and-requirements-of-dns-for-ioa-01

Example use case:
- Expedia booking agent prompt (seek other property agents i.e. Marriott)
- Expedia searching for properties (seek properties i.e. rooms/costs)
- User selection, Expedia agent interfaces with specific property (transaction)

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Core Concepts: Problems

1. Ephemeral (dynamic de/registration), workload-specific agents could scale to billions. Discovery requests orders of magnitude more.

2. Context-aware trusted discovery of offered services and cost with integration to public/private networks with operators controlling advertising their services.

3. Legal dispute for agent removal or unauthorized access, legislation enforcing data sovereignty, and enforcing uniqueness / anti-collision mechanisms globally.

Solution Space

Two main proposals, ANS and BANDAID. We agree... DNS w/ SVCB = 👑

Drafts:
- draft-mozleywilliams-dnsop-bandaid-00
- draft-narajala-ans-00
- draft-zyyhl-agent-networks-framework-01

ANS proposed DNS-like mechanisms for resolution of service/location in a new proprietary protocol. BANDAID proposed a leaf attribute zone (i.e. _agents.example.com) utilizing SVCB records with new parameters (cost, input token amounts, etc.) with no changes to existing DNS protocol.

There is rough consensus between author groups SVCB provides more flexibility and reliability in an existing protocol as an entry point for discovery.

Solution High Level Overview

Known entry point →
SVCB/HTTPS lookup →
DNSSEC validate →
DCV check (if required) →
DANE/TLSA pin (optional) →
Connect using advertised ALPN/port →
Monitor/audit

???

Profit.

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Solution Features

Publish agents under a structured leaf zone (i.e. _a2a._agents.example.com) with SVCB/HTTPS records that advertise endpoints, protocols, capability locators, cost metadata, etc.

Anchor trust and global uniqueness via DNSSEC signed zones with full DS chain, bind transport identity with DANE (TLSA)... agents validate authenticity before protocol handshakes.

Prove delegated authority using DCV: short-lived DNS TXT challenges and role/scope assertions signal an agent is authorized to act for a domain, supporting ephemeral credentials and least-privilege access.

Preserve operator sovereignty and policy control with split-horizon DNS (internal vs. external views), enterprise controls (e.g., RPZ/ACLs), and hierarchical delegation for multitenant and federated environments.

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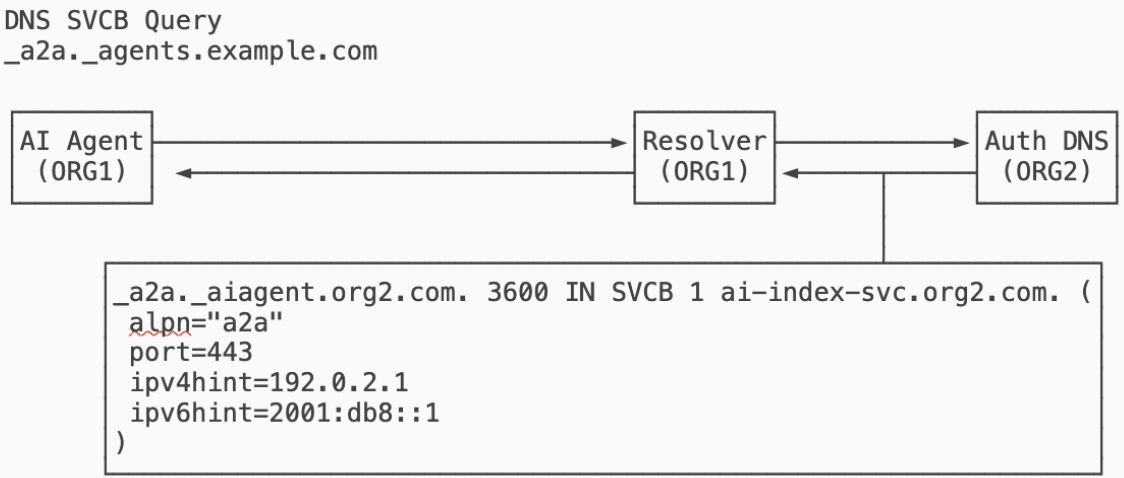
Solution Features (cont.)

Manage ephemeral agents cleanly with per-service/per-agent leaves, right-sized TTLs, **IXFR/NOTIFY** for rapid updates, and operational telemetry for lifecycle, reliability, and cost insights. Performance scaling by leveraging DNS's caching, local-root priming, prefetch/serve-stale, anycast authoritative and distributed resolvers, SVCB address hints, and EDNS(0)/TCP fallback.

Interoperate on both public and private networks and across agent protocols (A2A, MCP, ACP). DNS-SD enables capability-based discovery; once found, agents connect using the advertised transports and application protocols.

Plan for schema evolution and economic signaling: standardize capability schemas and use BANDAID SVCB parameters (e.g., capability references, app protocol list, policy realm) to include cost/capability metadata with versioning.

Diagram: "It's Just DNS!"



SVCB params to be modified to include application-specific context (input token amounts, cost per input bundle, etc.)

Example: Zone File + SVCB Record

- Use SVCB ServiceMode (priority > 0) to bind connection parameters.
- AliasMode to keep a friendly name stable even if the leaf changes (where **a4k2f9** is the hashed identifier of the specific agent as it's republished).

$ORIGIN example.com. $TTL 3600

; @ IN SOA ns1.example.com. hostmaster.example.com. (2025091901 ; serial (YYYYMMDDnn) 7200 ; refresh 1800 ; retry 1209600 ; expire 3600) ; minimum IN NS ns1.example.com. IN NS ns2.example.com.

ns1 IN A 192.0.2.53 ns1 IN AAAA 2001:db8::53 ns2 IN A 192.0.2.54 ns2 IN AAAA 2001:db8::54

billing._mcp._bandaid 300 IN SVCB 0 **a4k2f9**._mcp._bandaid.example.com.

Example: Leaf Service

- Leaf per agent/service (ServiceMode). Shorter TTL for agility.
- Gate client requirements with "mandatory".

```
;  a4k2f9._mcp._bandaid 600 IN SVCB 1 svc-a4k2f9.example.com. \
      alpn="h2,h3" \
      port=443 \
      ipv4hint=192.0.2.5 \
      ipv6hint=2001:db8::5 \
      mandatory=alpn,port
```

Example: SVCB Extensibility

- Capability / policy signaling via SVCB custom keys (experimental).
- Example: reference a capability descriptor and advertise supported app protocols.

```
; a4k2f9._mcp._bandaid 600 IN SVCB 1 svc-a4k2f9.example.com. \
    alpn="h2,h3" port=443 \
    mandatory=alpn,port,key65001,key65010 \
    key65001="cap=urn:cap:example:mcp:invoice.v1" \
    key65010="bap=a2a/1,mcp/1"
```

Example: DNS-based Authentication + Authorization

- Domain Control Validation (DCV) for BANDAID authorization.
- Publish a short-lived token to prove control over example.com.
- Remove after successful validation.
- Resolver MUST DNSSEC-validate.

; _bandaid-challenge 300 IN TXT "bnd-req=svc:crm-sync@vendor.tld;nonce=3Qz6l8pA;exp=2025-09-19T06:00:00Z"

Example: Bind TLS Identity to DNS

- Optional DANE TLSA for the service endpoint used above.
  - Owner: _<port>._tcp.<endpoint-FQDN>
  - Usage 3(DANE-EE), Selector 1 (SPKI), Match 1 (SHA-256)
- Replace the hash with the actual SPKI hash of the leaf certificate.

```
;  _443._tcp.svc-a4k2f9.example.com. 1800 IN TLSA 3 1 1
( 0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF0123456789AB )
```

Security Considerations

Lookalikes / domain reputation. Enforcing connections to marriott.com (2 T's) instead of marriottt.com (3 T's) despite fluctuations in agent temperature.

Adversarial use of agents (using inputs from domains to modify an agent's reasoning).

IANA Considerations

- What SVCB params *SHOULD* be defined for BANDAID?
  - Cost, input tokens, capabilities, policy realm, app protocols, versioning, latency, location, real-time versus short or long term run times, misc. qualifications
- What leaf zone(s) *SHOULD* be defined for BANDAID?
  - See draft-liu-cats-dns-service-discovery-01 where a directory service is proposed to advertise computational capabilities: cpu, mem, lat, load, gpu, accel, vers, caps, prio, cost, avail within DNS leaf zones for each
    - We argue this is needlessly complex and SRV records get unwieldy quick with this setup
    - Should we use this already formed WG to discuss? Agent to agent the best venue?

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"The Debate"

- Leaf attribute zones for specific capabilities (i.e. _chat._agents.example.com)?
  - Or, better for a protocol to have agents within and records hosting SVCB with capabilities in _agents.example.com where booking._agents.example.com includes ALPN or proto = A2A in the record?
    - These need to be registered with IANA. What if you do something that isn't yet defined like chat/booking? Delays in processing...
    - What if you don't know what you are looking for? If you want to see all agents, does that lend credibility to an index-like service where there are well-known agent capabilities registered with IANA and innovative ones self hosted?
- AI agents typically use an API call to Google Search, or screen capture a browser UI and analyze the picture. They favor HTTPS transactions.
- How do we make the connection between 'brand' to 'domain' in a predictable manner?
  - Reputation weighting of domains or nameservers?

Calls for Collaboration

Jim presented at DISPATCH this morning. There is rough consensus the next step for this scope is WG-forming BOF.

Please help!

Discussion / Q&A